

Atmospheric Sciences 528: Atmospheric Data Analysis
 Dr. Jared Marquis (Fall 2026)
 Assignment #1: The Polar Stereographic Projection
 Due: 11 September 2026 by 11:59PM
 50 pts

1. Show that the radius of a latitude circle on the image plane r is equal to

$$r = \rho(1 + \cos \psi_0) \tan\left(\frac{\psi}{2}\right)$$

2. Show that

$$\sigma = \frac{\rho(1 + \cos \psi_0) \tan\left(\frac{\psi}{2}\right)}{\rho \cos \varphi}$$

can be written as

$$\sigma = \frac{1 + \sin(\varphi_0)}{1 + \sin(\varphi)}$$

3. Show that the latitude and longitude of a point (x,y) on the image plane can be calculated according to

$$\varphi = \frac{\pi}{2} - 2 \tan^{-1} \left\{ \frac{(x^2 + y^2)^{\frac{1}{2}}}{\rho[1 + \cos(\psi_0)]} \right\}$$

$$\lambda - \lambda_0 = \tan^{-1} \left(\frac{y}{x} \right)$$

4. Suppose that you wish to construct a polar stereographic map of the entire northern hemisphere utilizing a piece of paper with dimensions 20 x 20 inches. What would be the largest map scale that you could use? Let $\varphi_0 = 60$ degrees.